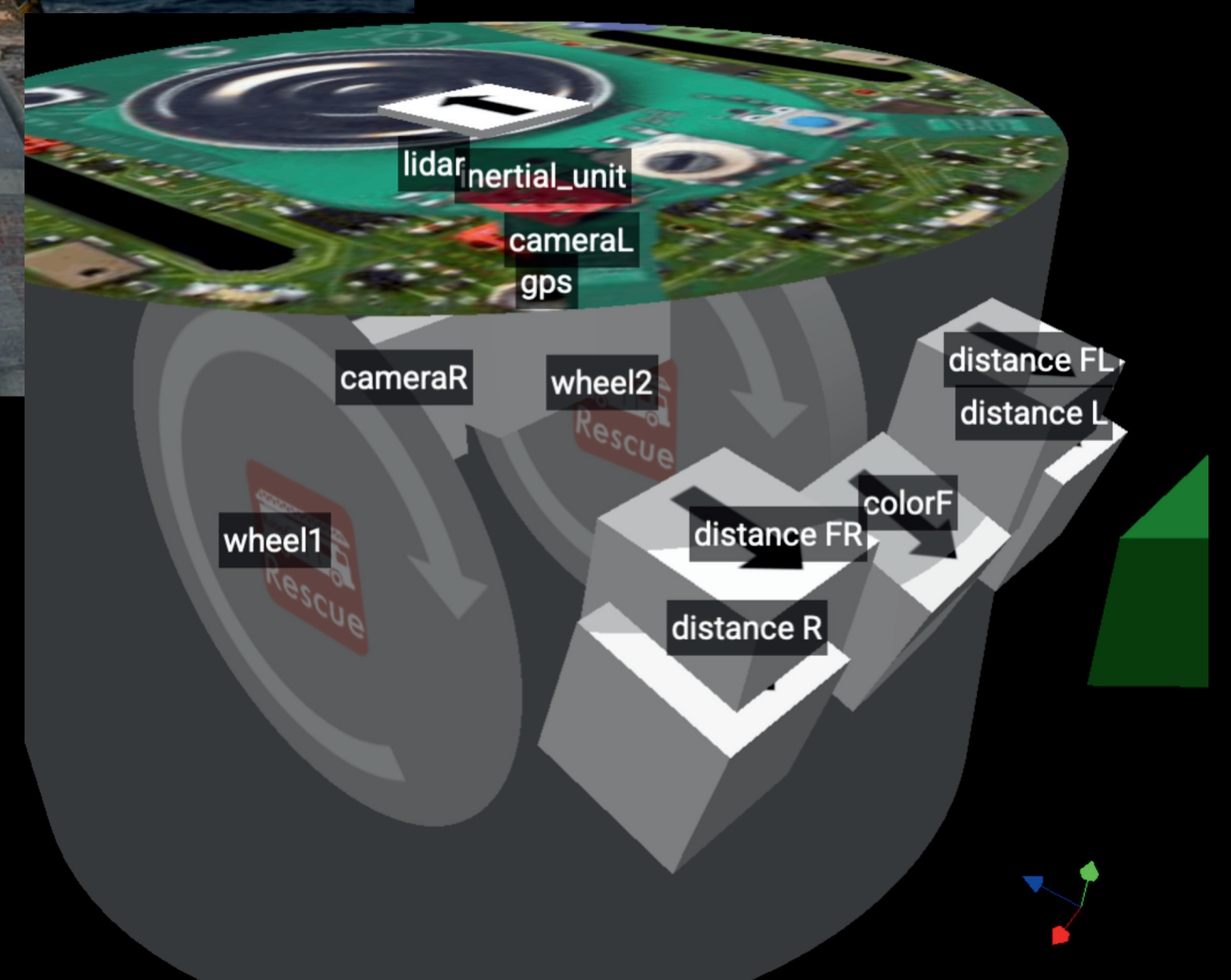


from Taiwan

- Chen Yu Wei (16y)
- Engineering Journal , coding
- Early exposure to programming
- arduino, Ev3,Vex v5 Domestic and international competitions

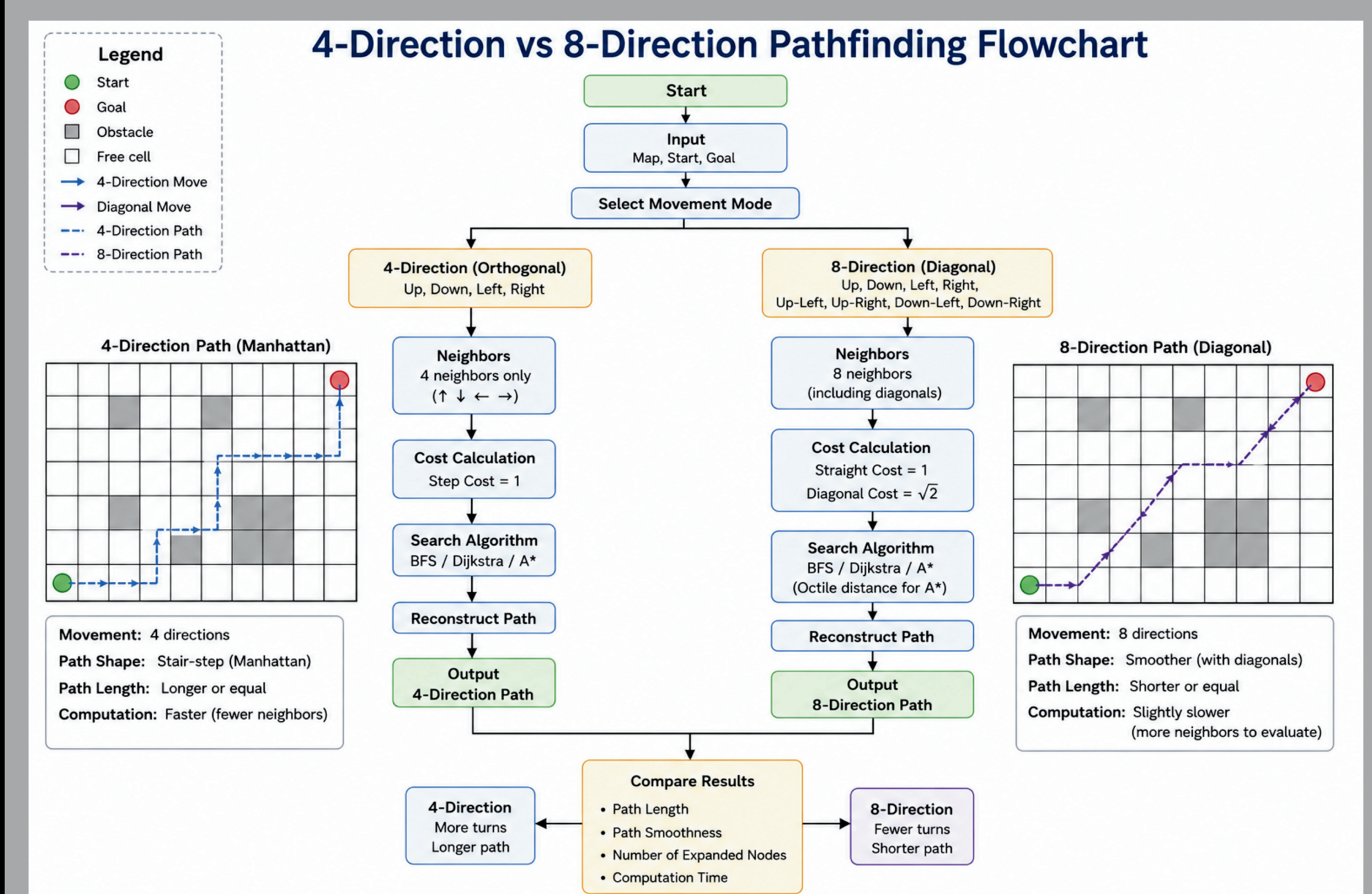
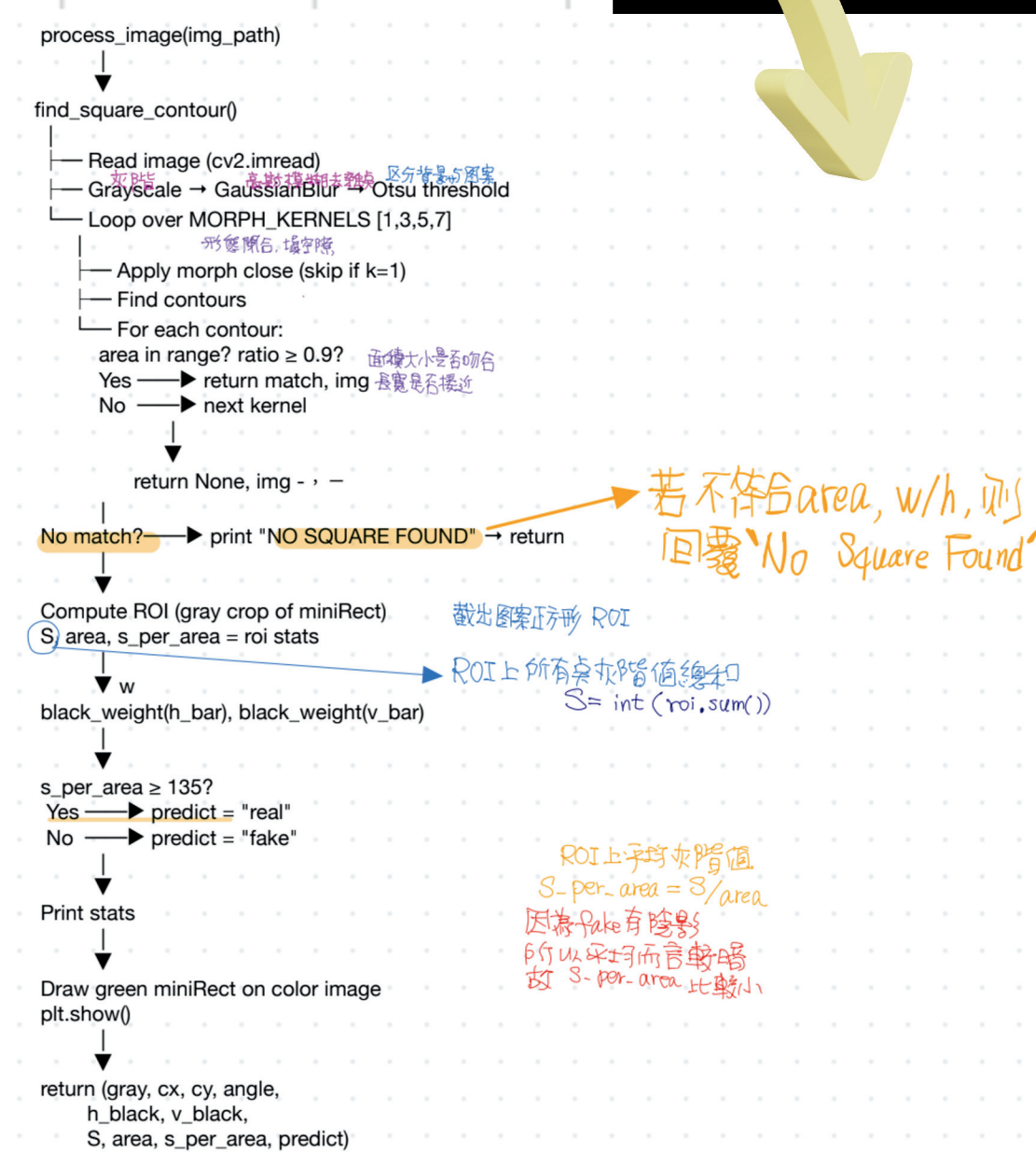
-
- A photograph of three young men standing outdoors in front of a large white ship and a city skyline. The man on the left is wearing a dark blue t-shirt and making a peace sign. The man in the middle is wearing a white t-shirt with 'WE11DONE' and also making a peace sign. The man on the right is wearing a light yellow t-shirt and giving a thumbs up. A large, stylized logo for 'RESCUE LEAGUE LABYRINTHOS' is overlaid in the foreground. The logo features a blue and orange color scheme with a star, a compass rose, and a robot head. The background shows a waterfront area with palm trees and a city skyline under a clear sky.



	A	B	C	D	E	F
1	filename	area	aspect_ratio	morph_k	angle	cx
2	omegaF1.jpg	372	0.917	3	-90	
3	omegaF2.jpg	576	1	1	-90	29
4	omegaF3.jpg	263.5	1	5	-90	42.5
5	omegaF4.jpg	332.5	1	5	-90	36
6	omegaF5.jpg	288	0.95	7	-90	18.5
7	phiF1.jpg	326	0.95	7	-90	53
8	phiF2.jpg	377	1	5	-90	34
9	phiF3.jpg	313	1	3	-90	15
10	phiF4.jpg	321	0.96	3	-90	41.5
11	psiF1.jpg	291	0.958	3	-90	13.5
12	psiF2.jpg	312	0.958	1	-90	26.5
13	psiF3.jpg	281	1	5	-90	49

- 4 Distances(F/R/FR/FL) :Distance Detection for Obstacle Avoidance
- 1 Color sensor : Floor color marker detection
- 2 Camara(R/F) : Object detection for scoring
- 1 Lidar :Enhancing positional accuracy by matching LiDAR scans against the known field layout
- 1 Inertial_unit : Maintaining a constant heading during autonomous navigation to prevent drifting
- 1 GPS:Geofencing & Area Marking

We found that grayscale values can be used to highlight colors.



A. Hazard Determination Logic

Total Score	Hazard Type	
0		Flammable Gas
1		Poison
2		Corrosive
3		Organic Peroxide
Other Scores		Fake Target

The diagram illustrates the five steps of the machine learning model for target detection:

- 1 Image Preprocessing:**
 - Input Image
 - Grayscale
 - Gaussian Blur
 - Binary (Inverse)
- 2 Contour Screening:**
 - Detect Contours
 - Filter (Area, Shape, Position)
 - Keep Target Region
- 3 Core Refinement:**
 - Check Center Offset
 - Update Inner Contour Center
- 4 Horizontal Color Line Scan:**
 - Scan Color Order
 - Compare Color Differences
 - Remove Noise
- 5 Real vs. Fake Decision:**
 - Real Target (True Label)
 - Fake Target (False Label)

The output of the process is fed into a **Machine Learning Model** to **Classify: Real Target vs. Fake Target**.

Test scores

Image recognition accuracy
95% up

Map Coverage Percentage
85% up

[illegible]